

Classical Mechanics — Study Notes

VERA Test Document · Physics · 4 pages

1. Newton's Laws of Motion

Newton's first law states that an object at rest remains at rest, and an object in motion continues in motion with constant velocity, unless acted upon by a net external force. This principle is called the law of inertia.

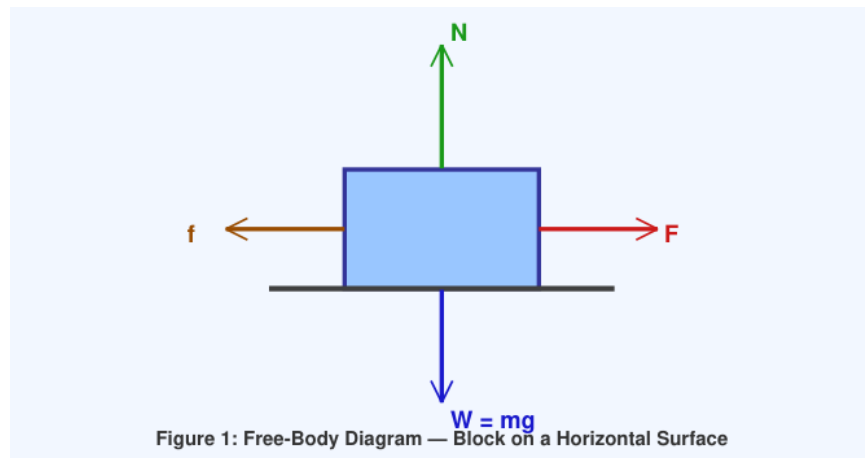
Newton's second law of motion states that the net force acting on an object equals the product of its mass and its acceleration. This is the central equation of classical mechanics:

$$F = ma$$

where F is net force in Newtons, m is mass in kilograms, and a is acceleration in metres per second squared. Newton's third law states that for every action there is an equal and opposite reaction.

1.1 Free-Body Diagrams

A free-body diagram isolates a single object and shows all forces acting on it. The diagram below shows a block on a horizontal surface with applied force F , normal force N , weight $W = mg$, and friction force f opposing motion.



For vertical equilibrium the normal force equals the weight: $N = mg$. Friction is related to the normal force by the coefficient of kinetic friction:

$$f = \mu_k N$$

2. Work, Energy, and Power

Work is the dot product of force and displacement. When a constant force F acts through displacement d at angle θ to the displacement:

$$W = Fd\cos\theta$$

The work-energy theorem states that net work done on an object equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

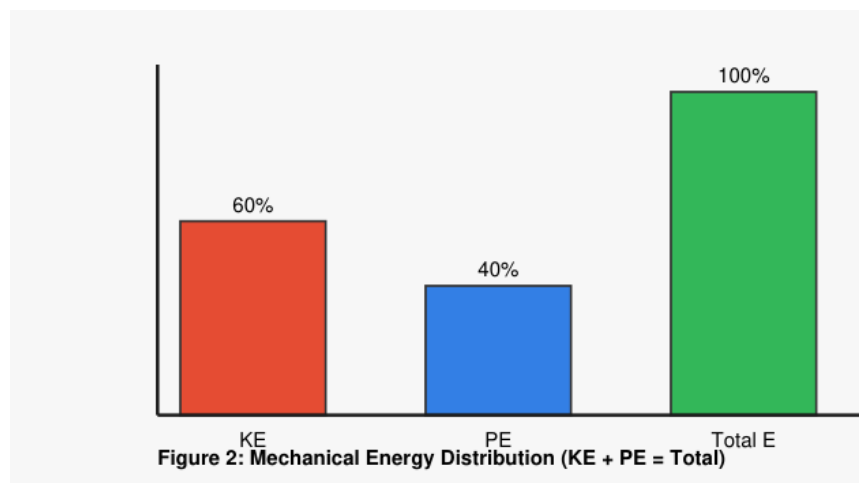
2.1 Kinetic and Potential Energy

The kinetic energy of an object of mass m moving at speed v is $KE = \frac{1}{2}mv^2$. Gravitational potential energy near Earth is $PE = mgh$ where h is height above a reference level and g equals 9.8 m/s^2 .

$$KE = \frac{1}{2}mv^2$$

$$PE = mgh$$

Conservation of mechanical energy: in the absence of friction the total energy $E = KE + PE$ remains constant. The diagram below shows how KE and PE split the total energy at a given instant:



2.2 Power

Power is the rate at which work is done. Average power equals work divided by time. The SI unit of power is the Watt, equal to one Joule per second.

$$P = \frac{W}{t} = Fv$$

3. Kinematics — Equations of Motion

Kinematics describes motion without reference to its causes. For uniform acceleration the four SUVAT equations apply:

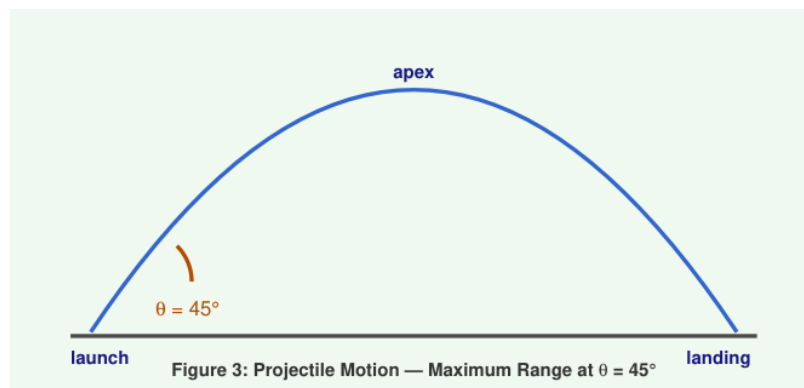
Equation	Links
$v = v_0 + at$	v, v_0, a, t
$s = v_0 t + \frac{1}{2}at^2$	s, v_0, a, t
$v^2 = v_0^2 + 2as$	v, v_0, a, s
$s = \frac{(v_0 + v)}{2} t$	s, v_0, v, t

3.1 Projectile Motion

Projectile motion is two-dimensional motion under constant gravitational acceleration. The horizontal velocity is constant; the vertical component changes due to gravity. The range of a projectile launched at angle θ with initial speed v_0 is:

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Maximum range is achieved at a launch angle of 45 degrees. The diagram below shows the parabolic trajectory at maximum-range conditions:



3.2 Circular Motion

Uniform circular motion requires centripetal acceleration directed toward the centre. Its magnitude is:

$$a_c = \frac{v^2}{r}$$

The centripetal force $F = mv^2/r$ is provided by tension, gravity, or normal force depending on the scenario.

4. Thermodynamics

Thermodynamics studies heat, work, and temperature in relation to energy and entropy. The four laws of thermodynamics govern all thermal processes.

4.1 First Law

The first law of thermodynamics states that the change in internal energy of a system equals the heat added minus the work done by the system:

$$\Delta U = Q - W$$

4.2 Carnot Efficiency

The second law of thermodynamics states that the total entropy of an isolated system always increases. No engine can exceed Carnot efficiency. The Carnot efficiency for a heat engine operating between temperatures T_C and T_H is:

$$\eta = 1 - \frac{T_C}{T_H}$$

where T_C is the cold reservoir temperature and T_H the hot reservoir temperature, both in Kelvin. This is the theoretical maximum for any heat engine.

4.3 Ideal Gas Law

The ideal gas law combines Boyle's law, Charles's law, and Avogadro's law:

$$PV = nRT$$

where P is pressure in Pascals, V is volume in cubic metres, n is moles, R is the universal gas constant 8.314 J per mol per Kelvin, and T is absolute temperature in Kelvin.

5. Key Physical Constants

Constant	Symbol	Value
Gravitational acceleration	g	9.8 m/s²
Universal gas constant	R	8.314 J/(mol·K)
Speed of light	c	3.0 × 10 ⁸ m/s
Planck constant	h	6.626 × 10 ⁻³⁴ J·s
Boltzmann constant	k_B	1.38 × 10 ⁻²³ J/K